

Smart Hydroponics System

User Manual

Version 1.0

1. System Overview

The Smart Hydroponics System is an automated plant cultivation platform designed to monitor and control nutrient delivery, water level, temperature, humidity, pH, and environmental conditions. The system helps maintain optimal growth conditions and can provide alerts when abnormal conditions are detected.

How It Works:

1. Fill the nutrient reservoir.
2. Power on the controller.
3. Sensors continuously monitor conditions.
4. The controller automatically operates pumps and monitoring functions.
5. Users can monitor system status through the display or connected interface.

2. What You Need

Hardware:

- Controller Unit
- Water Pump
- Nutrient Reservoir
- pH Sensor
- Temperature & Humidity Sensor
- Water Level Sensor
- Power Supply

Software:

- Monitoring Dashboard
- Mobile/PC Browser

Accounts:

- Wi-Fi Network (if applicable)
- User Account for Remote Monitoring

3. Initial Setup

- Step 1 – Assemble the hydroponic setup.
- Step 2 – Connect all sensors and pumps.
- Step 3 – Fill the reservoir with water and nutrients.
- Step 4 – Power on the system.
- Step 5 – Verify sensor readings and pump operation.

4. How to Use the System

1. Start Monitoring.
2. Check water level and nutrient concentration.
3. Observe temperature and humidity values.
4. Review alerts and notifications.

5. Perform periodic maintenance.

5. System States

IDLE – System powered but inactive.

MONITORING – Sensors actively collecting data.

IRRIGATION ACTIVE – Pump operating.

ALERT MODE – Abnormal condition detected.

6. Monitoring Your Hydroponics System

The dashboard displays:

- Water level
- Temperature
- Humidity
- pH value
- Pump status
- System health indicators

7. Control Commands

START – Begin monitoring.

STOP – Stop monitoring.

STATUS – Display current readings.

RESET – Restart the controller.

8. Troubleshooting

Case 1: Pump not operating.

- Check power supply.
- Verify wiring.

Case 2: Incorrect sensor readings.

- Recalibrate sensors.
- Check sensor connections.

Case 3: Low water level alert.

- Refill reservoir.

Case 4: pH out of range.

- Adjust nutrient solution.

9. Technical Specifications

Power Supply: 12V DC

Water Pump: Automated Control

Sensors: pH, Water Level, Temperature, Humidity

Communication: Wi-Fi / Serial Interface

Display: LCD/OLED Dashboard

Operating Environment: Indoor Agriculture

Safety and Best Practices

- Keep electrical components dry.
- Clean sensors periodically.
- Replace nutrient solution regularly.
- Ensure adequate ventilation.
- Inspect pump operation weekly.

Maintenance

- Clean reservoir every growth cycle.
- Verify sensor calibration monthly.
- Inspect tubing and pump connections.
- Backup monitoring data when required.

Quick Reference

START → Start Monitoring

STOP → Stop Monitoring

STATUS → View Current Status

RESET → Restart Controller